Lecture -Tutorial 2

MAT700 - Mathematical Methods for Data Mining,



Jaccard Similarity

Distance Measures

k-Nearest Neighbour or k-NN classification

Model Evaluation

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Problem 1.

Compute the Jaccard similarities of the following two bags: $\{1,2,2,3,3,3,4,4,4,4\}$ and $\{4,3,3,2,2,2,1,1,1,1\}$.

Distance Measures



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Problem 1.

On the space of non-negative integers, which of the following functions are distance measures?
If so, prove it; if not, prove that it fails to satisfy one or more of the axioms.

(a) $\max(x, y)$ = the larger of x and y.

(b) $diff(x, y) = |x - y|$ (the absolute magnitude of the difference between x and y).

(c) $sum(x, y) = x + y$.

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Problem 2.

Find the L_1 and L_2 and L_∞ distances between the points $(5, 6, 7)$ and $(8, 2, 4)$.

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Problem 3.

Find the Hamming distances between each pair of the following vectors: 000000, 110011, 010101, and 011100.

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Problem 3.

Find the Hamming distances between each pair of the following vectors: 000000, 110011, 010101, and 011100.

Problem 4.

Prove that the L_∞ distance is indeed a distance measure.

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Problem 5.

Prove that the **edit distance** is indeed a distance measure.

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Problem 5.

Prove that the **edit distance** is indeed a distance measure.

A Longest Common Subsequence

Another way to define and calculate the edit distance $d(x, y)$ is to compute a *longest common subsequence* (**LCS**) of x and y. An LCS of x and y is a string that is constructed by deleting positions from x and y, and that is as long as any string that can be constructed that way. The edit distance $d(x, y)$ can be calculated as the length of x plus the length of y minus twice the length of their LCS.

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Problem 5.

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Example

Consider $x = aba$ and $y = bab$. Their edit distance is 2. There are two LCS's: *ab* and *ba*. Each can be obtained by deleting one symbol from each string. As must be the case for multiple LCS's of the same pair of strings, both LCS's have the same length.
Therefore, we may compute the edit distance as $3 + 3 - 2 \cdot 2 = 2$.

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Problem 1.

Consider the one-dimensional data set shown in Table

x	1.5	2.5	3.5	4.5	5.0	5.5	5.75	6.5	7.5	10.5
y	+	+	−	−	−	?	+	−	+	+

(a) Compute the class of $x = 5.5$ according to its 1- and 3-nearest neighbours (using majority vote).

(b) Repeat the previous exercise, but use the weighted version of kNN.

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Problem 2.

Given the following data:

instance	x_1	x_2	class
1	0.25	0.25	+
2	0.25	0.75	+
3	0.50	0.25	-
4	0.50	0.75	-
5	0.75	0.50	-
6	0.75	1.00	+
7	0.25	0.80	?

How does the weighted version of kNN algorithm classify instance 7, with $k = 4$ and L_∞ distance?

Model Evaluation

		Actual Class (observation) +	Actual Class (observation) -
Predicted Class (expectation)	+	**tp** (true positive) Correct result	**fp** (false positive) Unexpected result
	-	**fn** (false negative) Missing result	**tn** (true negative) Correct absence of result

$$\text{Precision} = \frac{tp}{tp + fp}$$

$$\text{Recall} = \frac{tp}{tp + fn}$$

$$F = 2 \cdot \frac{\text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}}$$

Problem 1.

During testing of a Spam/NotSpam classifier on a test set of 1000 e-mail messages with 100 spams and 900 not spams you found that 90 spams were correctly classified as spam, and 700 not spams were correctly classified as not spam. Calculate the precision, recall and F-measure of this classifier.

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